

LESSON 3-5 • SESSION 1 • PRACTICE • FORM A

Comparing Ratios with Common Terms

Learning target – I can compare two ratios by scaling them until they share a common term. • 6.RP.A.3b

REMEMBER

- To compare two ratios, **scale** them until one term is the **same** in both. Use a common multiple.
- Then compare the **other** term. Less water for the same soap means a **stronger** mix.
- Never compare the raw numbers: $3 : 7$ vs $2 : 5$ is not settled by $7 > 5$.

WORKED EXAMPLE

Mix A has 2 soap to 5 water. Mix B has 3 soap to 7 water. Which is stronger?

Step 1 Common multiple of 2 and 3: 6.

Step 2 $\text{Mix A} \times 3 = 6 : 15$. $\text{Mix B} \times 2 = 6 : 14$.

Step 3 Compare water: $14 < 15$.

Answer Mix B is stronger (less water for 6 soap)

WORDS common multiple (*múltiplo común*) • least common multiple (*mínimo común múltiplo*) • stronger / weaker (*más fuerte / más débil*)

PRACTICE

show your work • label units

1. Scale each ratio so the first term is 12.

a. $3 : 8 = 12 : \underline{\hspace{2cm}}$

b. $2 : 5 = 12 : \underline{\hspace{2cm}}$

c. $4 : 9 = 12 : \underline{\hspace{2cm}}$

d. $6 : 7 = 12 : \underline{\hspace{2cm}}$

2. **Purple Paint.** Mix A uses 3 cups red to 4 cups blue. Mix B uses 2 cups red to 3 cups blue. Complete the table. Which mix is bluer?

Hint Make the red 6 cups in both mixes.

	Red (cups)	Blue (cups)
Mix A $\times 2$	6	
Mix B $\times 3$	6	

Bluer mix _____

3. **Orange Juice.** Pitcher 1 has 3 cans of juice to 5 cans of water. Pitcher 2 has 2 cans of juice to 3 cans of water.

Hint Scale both pitchers to 6 cans of juice.

a. Scale both to 6 cans: _____

b. Which is more orange? _____

Show your work.

4. **Plant Food.** Bottle M mixes 5 drops of plant food with 9 ounces of water. Bottle N mixes 4 drops with 7 ounces.

a. Scale both to 20 drops: _____

b. Which is stronger? _____

Show your work.

5. **Hot Cocoa.** Cup A has 2 scoops of cocoa in 6 oz of milk. Cup B has 3 scoops in 8 oz. Scale both to 6 scoops. Which cup is more chocolatey?

Hint $A \times 3$ and $B \times 2$.

Show your work.

Cup _____

6. **Pool Chlorine.** Pool 1 uses 5 tablets for 10,000 gallons. Pool 2 uses 3 tablets for 5,000 gallons. Which pool has more chlorine per gallon?

Hint Scale Pool 2 up to 10,000 gallons.

Show your work.

Pool _____

7. **Bubble Solution.** Jar 1: 3 spoons of soap to 7 cups of water. Jar 2: 5 spoons to 12 cups. Scale both to 15 spoons of soap. Which is stronger?

Show your work.

Jar _____

8. **Team Wins.** Team 1 wins 4 of every 9 games. Team 2 wins 5 of every 12 games.

a. Scale both to 20 wins: _____

b. Which team wins a larger share of its games? Why? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Priya's Work.** Mix A: 2 cups white to 5 cups green. Mix B: 3 cups white to 7 cups green. Priya says B is greener because $7 > 5$. Explain her mistake. Which mix is greener?

TEST PRACTICE

choose the best answer

10. To compare $2 : 3$ and $3 : 5$ fairly, you first make –

A the sums equal **B** one term equal **C** both terms 0 **D** the totals 10

11. Which cocoa mix is stronger: 1 scoop to 3 cups of milk, or 2 scoops to 5 cups of milk?

A 2 to 5 **B** 1 to 3 **C** the same **D** can't tell

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

14 O	17 X	6 : 10 M	32 C	15 : 36 O	20 : 24 Z	20 : 45 N	20 : 36 M	6 : 5 Q
		#1a	#1d	#3a	#4a	#7	#8a	

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____